Attendance, Leave and Work Allocation Platform

Domain	Human Resource and Staff Administration
Problem Statement	Attendance, Leave and Work Allocation Platform
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1 Preface

- Traditional attendance, leave, and work management processes often rely on manual tracking and fragmented systems, leading to errors, delays, lack of transparency, and increased administrative effort.
- The Attendance, Leave, and Work Management Platform
- provides a centralized and secure solution to automate attendance tracking, leave application and approval, and work assignment management. The system improves operational efficiency, ensures data accuracy, and supports role-based access through an intuitive web interface.
- This document is intended for employees, team leads and HR who interact with the platform for daily workforce management activities and decision-making.

2 Introduction

2.1 Purpose of the System

- To automate and streamline employee attendance tracking, leave management, and work assignment processes**.**
- To improve efficiency, accuracy, and transparency in organizational workforce management.

2.2 Need for the System

- To provide a centralized platform for managing attendance, leave requests, and work allocation in an organized manner.
- Manual and disconnected systems often result in inaccurate records, delayed approvals, and increased administrative workload.

2.3 Overview of System Functions

- To enable employees to mark attendance, apply for leave, and view assigned work details through a unified interface.
- To simplify the work of team leads by enabling easy monitoring of team attendance, work progress, and leave approvals through a single platform.
- To enable HR to manage attendance and leave records, monitor workforce data, and generate reports for organizational needs.

2.4 Interaction with Other Systems

- The system interacts with a centralized database to store and manage employee details, attendance records, leave information, and work assignments.
- It can integrate with existing organizational systems such as payroll or reporting tools to ensure accurate data sharing and consistency.

2.5 Objectives of the System

- To develop an efficient and reliable system for managing employee attendance, leave requests, and work assignments.
- To simplify workforce management by enabling employees, team leads, and HR personnel to perform their tasks through a single integrated platform.
- To maintain accurate digital records of attendance, leave, and work data for easy access, monitoring, and reporting.

3 Glossary

Term	Meaning
Attendance	The record of an employee's daily presence, including check-in and check-out times.
Leave	Approved time off taken by an employee from work for a specific reason.
Task	A piece of work assigned to an employee with a defined objective.
Report	A summarized view of attendance, leave, or work details generated by HR or Team Lead.

4 User Requirements

- The system allows employees to mark daily attendance and view their attendance records online.
- The system enables employees to apply for leave and view leave status and available leave balance.
- The system allows team leads to view team attendance, approve or reject leave requests, and assign work to employees.
- The system enables employees to view assigned tasks and update work progress.
- The system allows HR to manage attendance and leave policies and generate attendance, leave, and work reports.

5 System Requirements

5.1 Functional Requirements

A. Access & Role Management

System provides role-based login and profile management for Employee, Team Lead, and HR.

B. Leave Management

Employees can apply for leave

Team Lead can approve or reject leave

System updates and maintains leave records

C. Work Allocation & Task Management

Team Lead can assign tasks to employees

Employees can update task status

Team Lead can monitor task progress

D. Attendance Management

Employees can mark daily attendance

System stores attendance records

Team Lead and HR can view attendance details

E. Reporting & Monitoring

System generates attendance, leave, and work reports

5.2 Non-Functional Requirements

Non-Functional requirements specify the quality attributes and constraints of the system such as performance, security, reliability, and usability.

A. Product Requirements

1. Efficiency Requirements

● The system should support multiple employees accessing the system simultaneously

● Leave, attendance, and work allocation operations should be processed efficiently

2. Performance Requirements

● Page response time should be less than or equal to 2 seconds

● Attendance marking and leave submission should be processed in real time

● Reports should be generated without noticeable delay

3. Space Requirements

● Efficient database storage should be used for employee, leave, and attendance records

● System should optimize storage for logs and reports

4. Usability Requirements

● System should have a simple and user-friendly interface

● Interface should be easy to use for employees, team lead, and HR

● System should be accessible through web browsers

5. Dependability Requirements

● System should be available during working hours without failure

● Data should be backed up regularly

● System should recover quickly from failures

6. Security Requirements

● Secure login and authentication mechanisms must be provided

● Role-based access control should be enforced

● Sensitive data such as personal details and attendance records should be protected

B. Organizational Requirements

1. Operational Requirements

● System should operate on web-based platforms

● System should support common operating systems and browsers

2. Environmental Requirements

● System should function properly under normal network conditions

● System should support use in office environments

3. Development Requirements

● System should be developed using standard programming frameworks

● Modular design should be followed for easy maintenance

● System should be scalable for future enhancements

C. External Requirements

1. Regulatory Requirements

● System should comply with organizational policies

● Employee data handling should follow data protection guidelines

2. Ethical Requirements

● Employee data should not be misused

● Attendance and leave data should be handled fairly

3. Legislative Requirements

● System should comply with IT laws and data privacy regulations

● Proper logs should be maintained for legal auditing

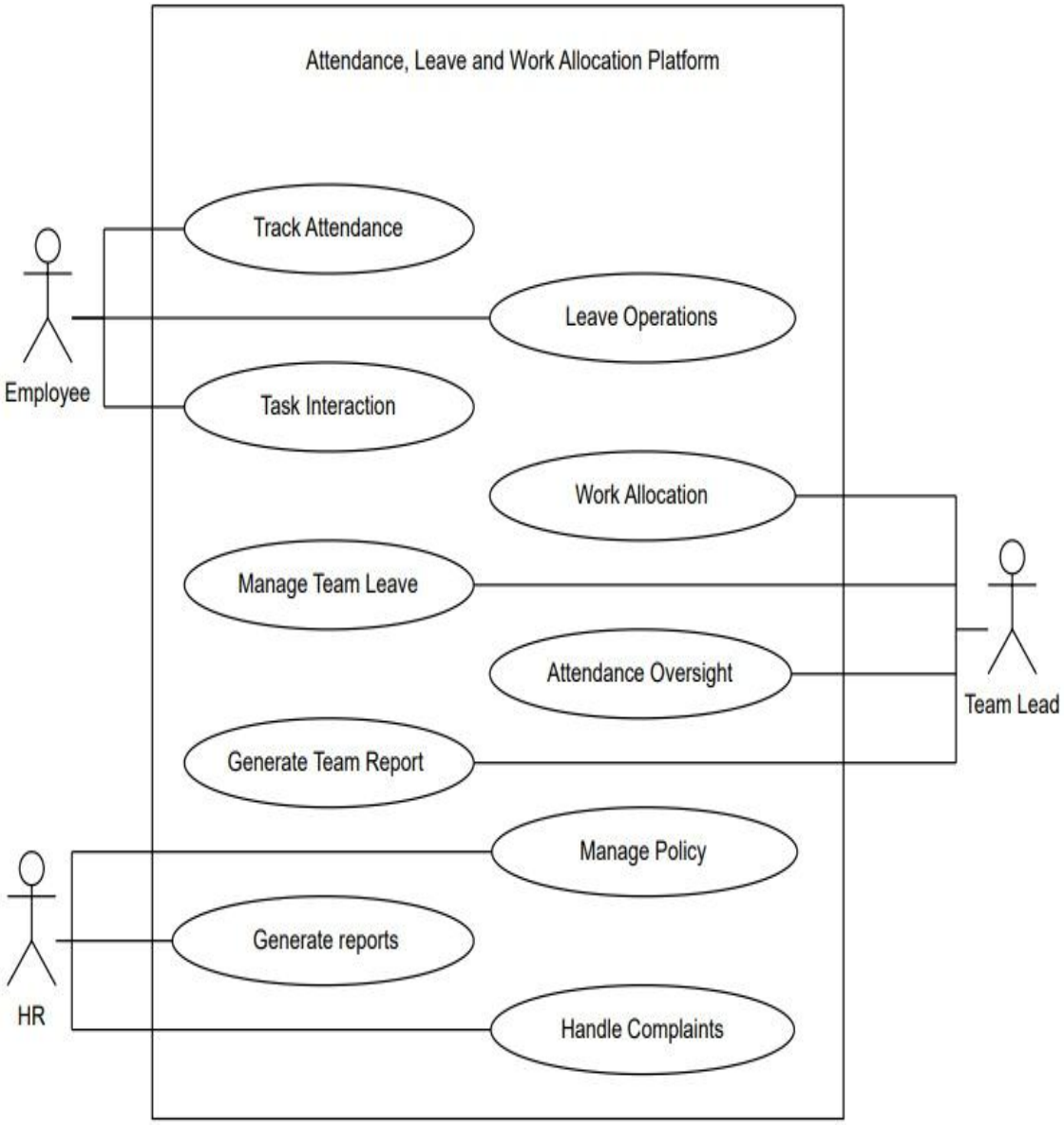
4. Safety / Security Requirements

● System should be protected from unauthorized access

● Measures should be taken to prevent data loss or misuse

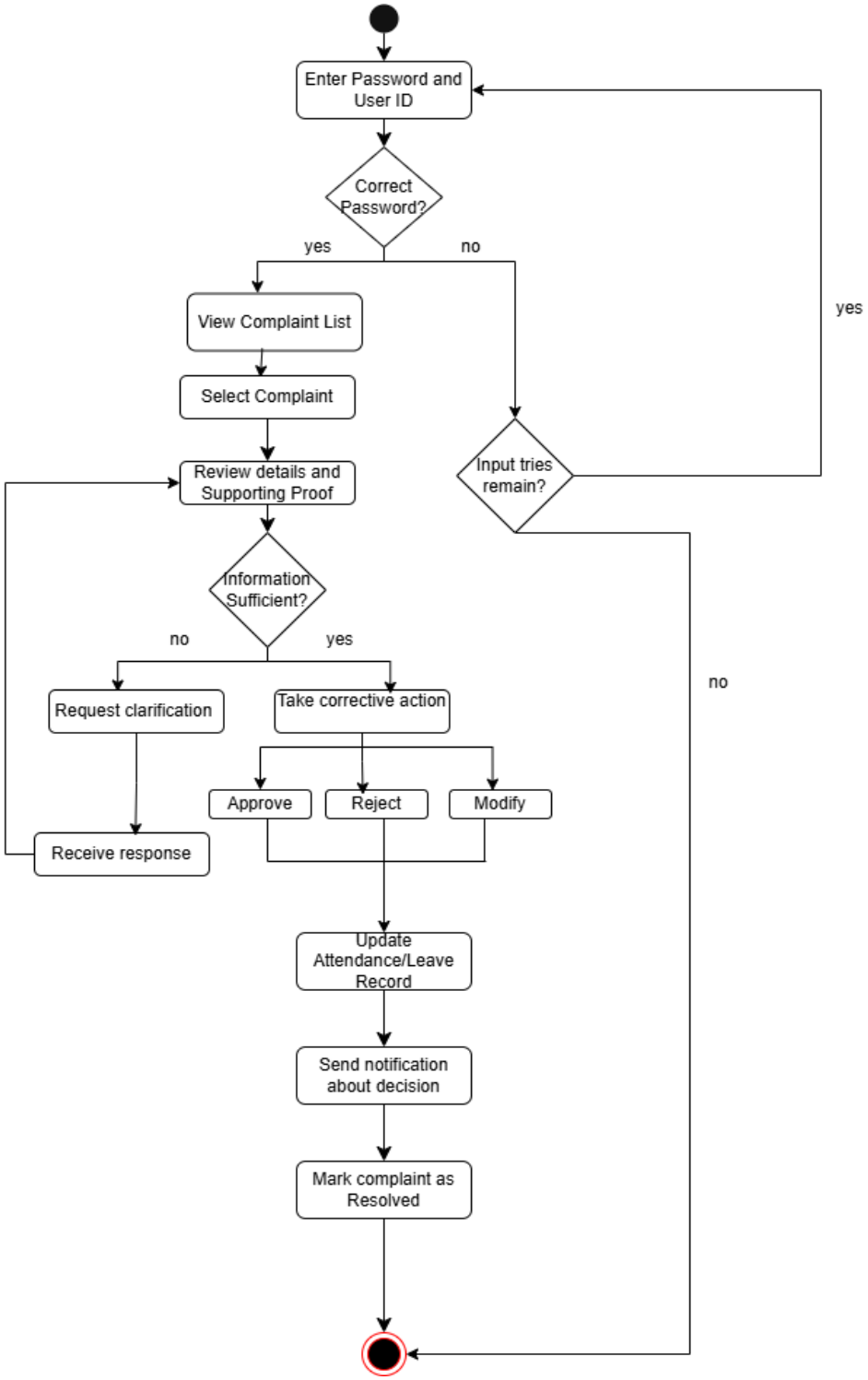
6 System Models

6.1 Use Case Diagram:

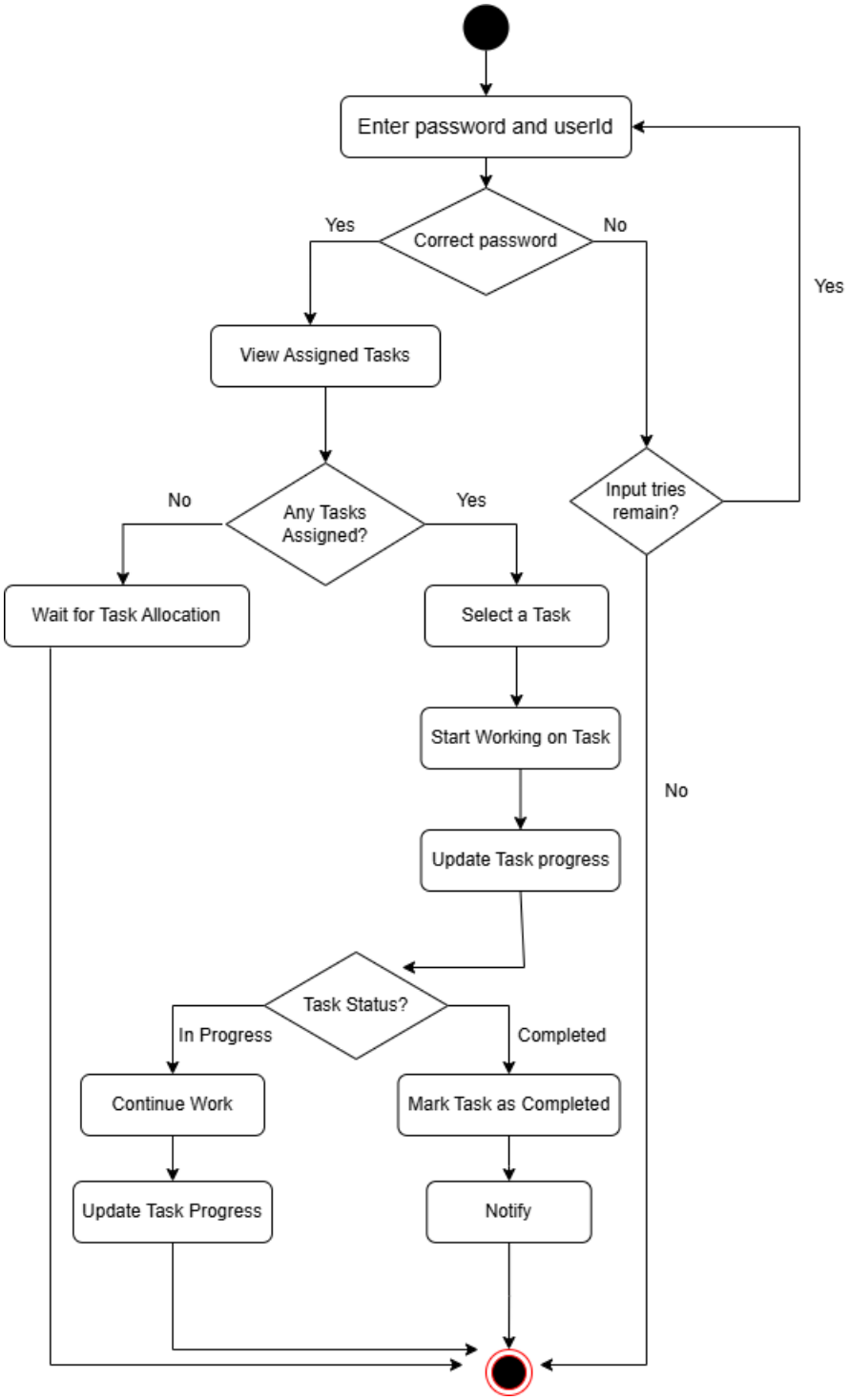


6.2 Activity Diagrams

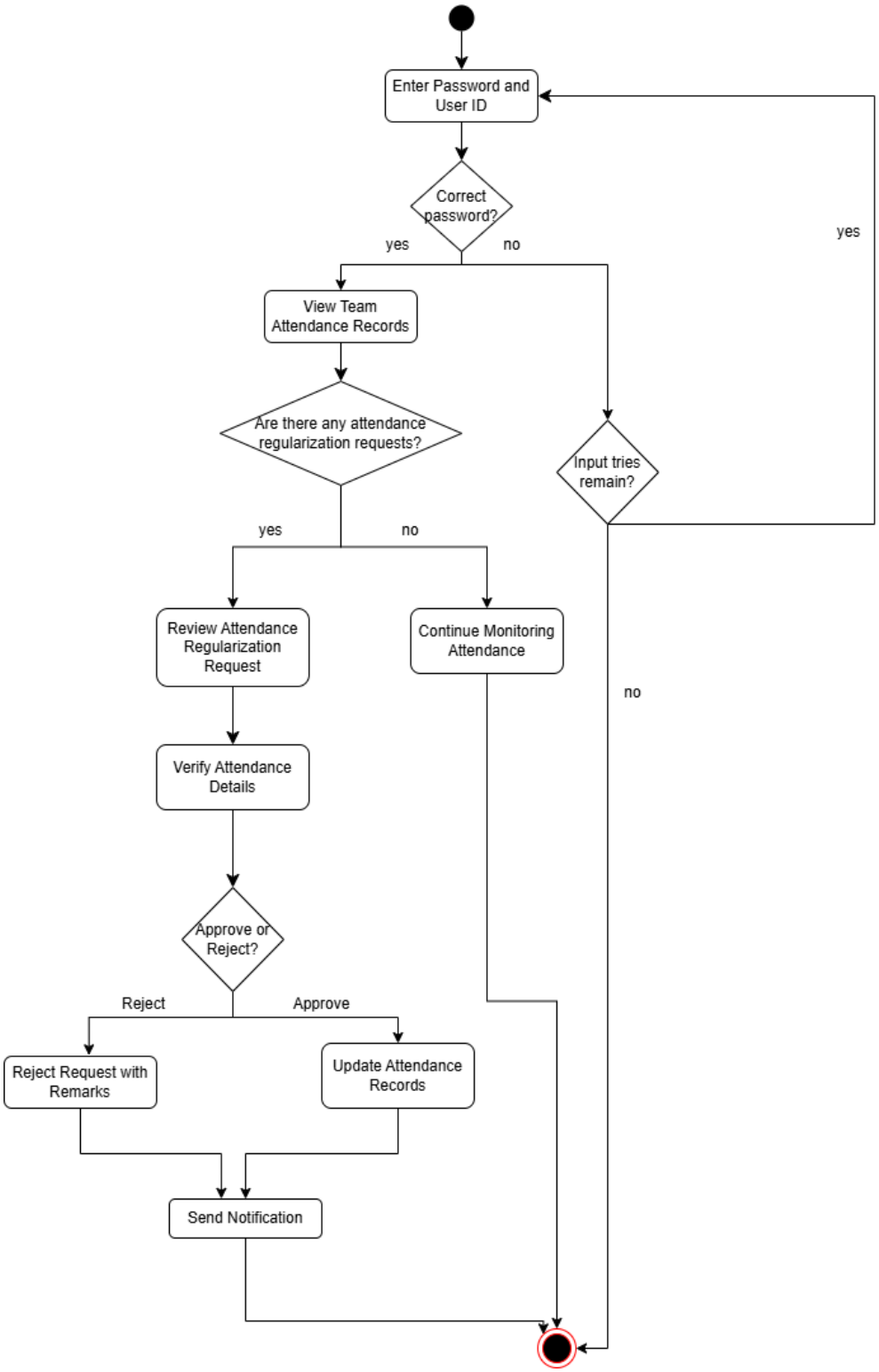
Handle Complaints:



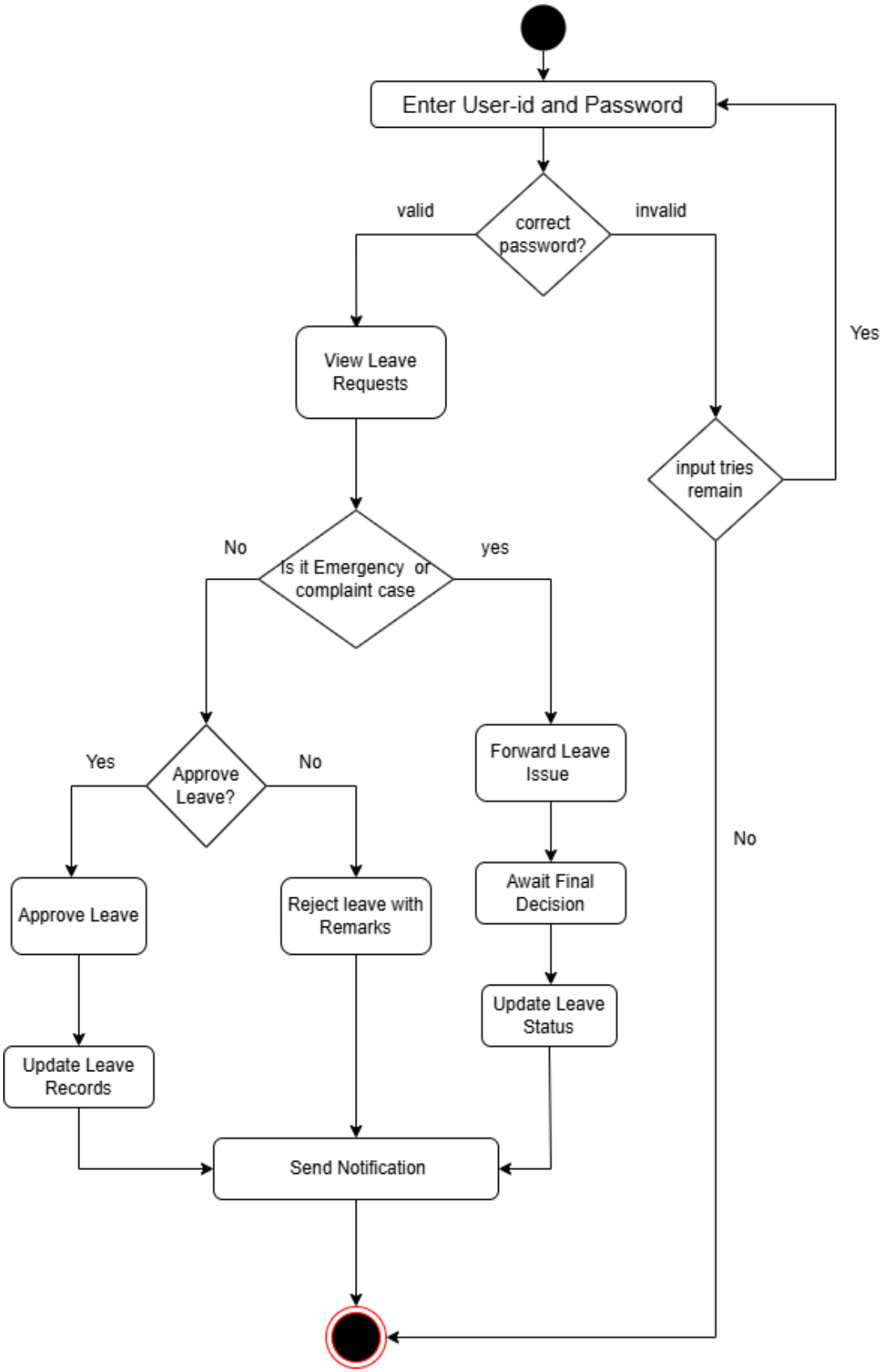
Task Interaction:



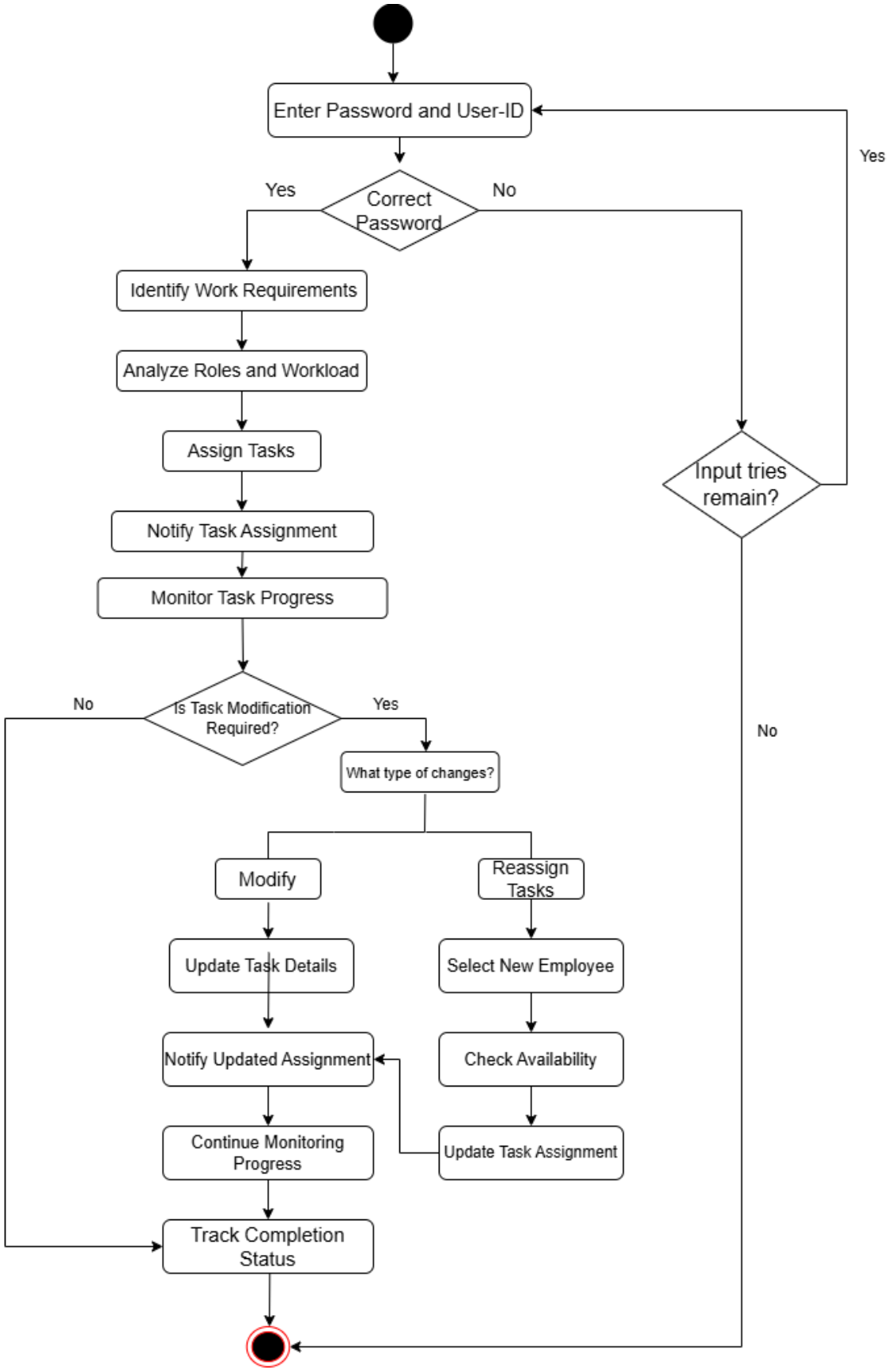
Attendance Oversight:



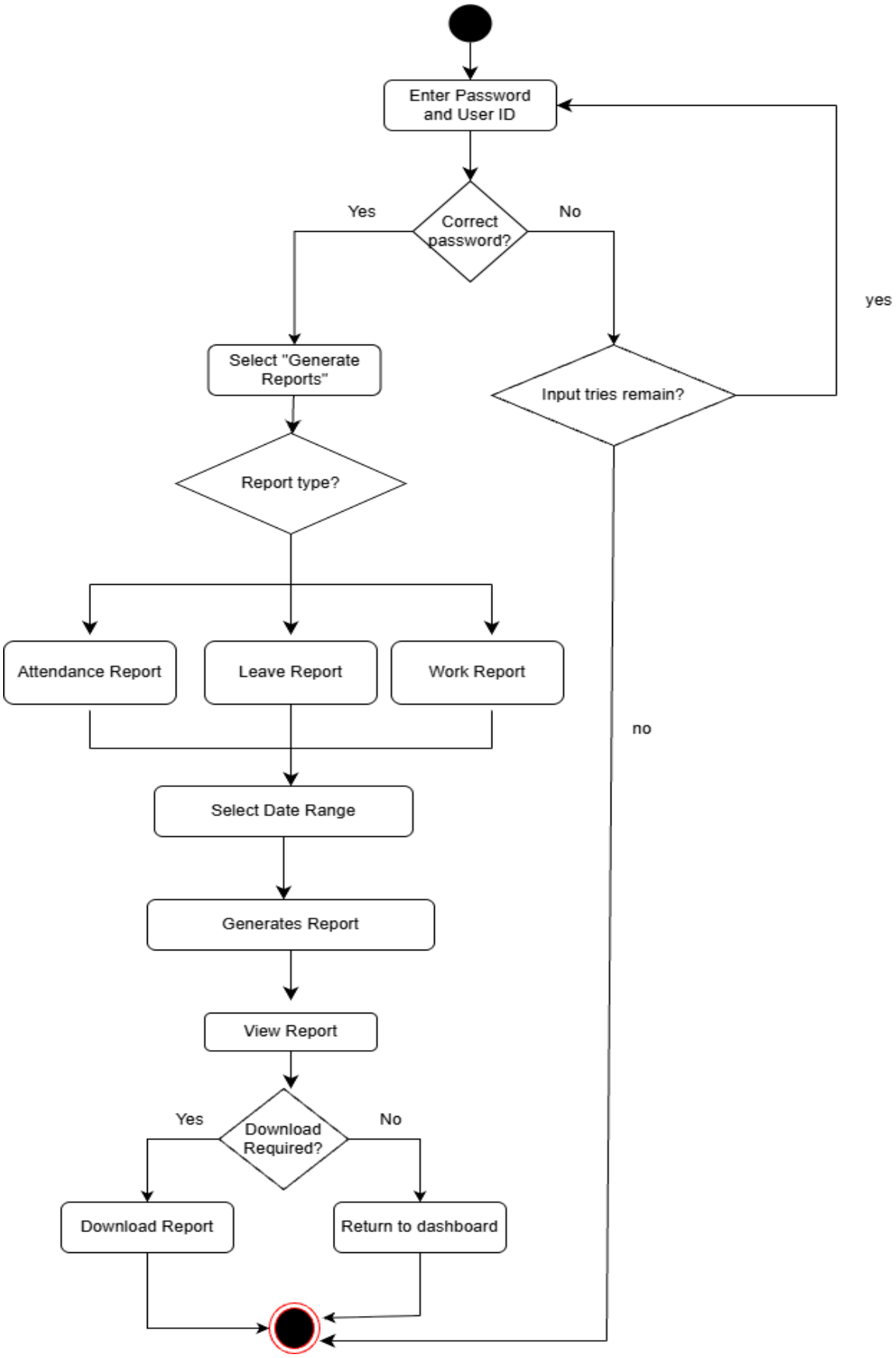
Manage Team Leave:



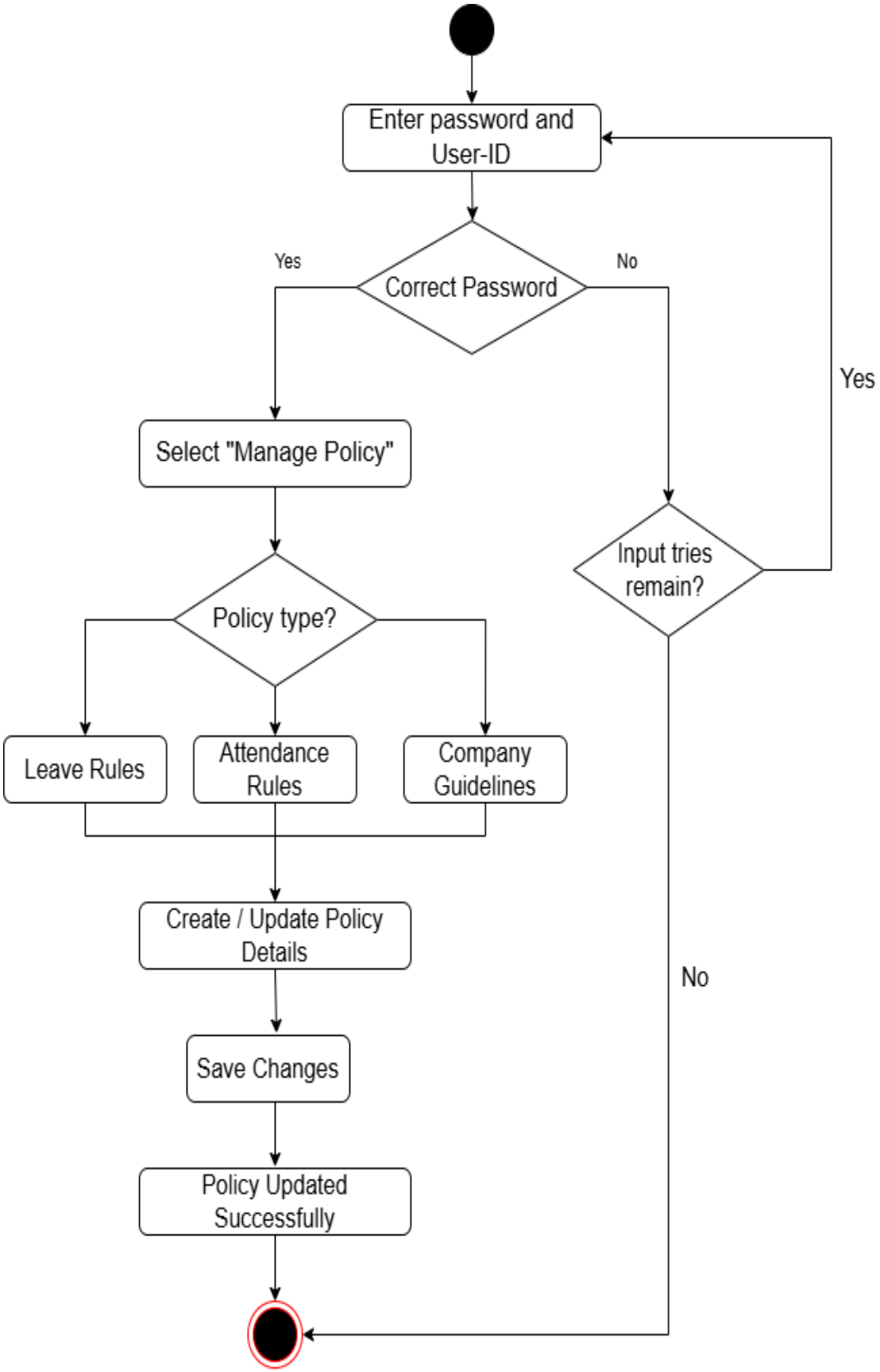
Assign Task Work:



Generate Reports:

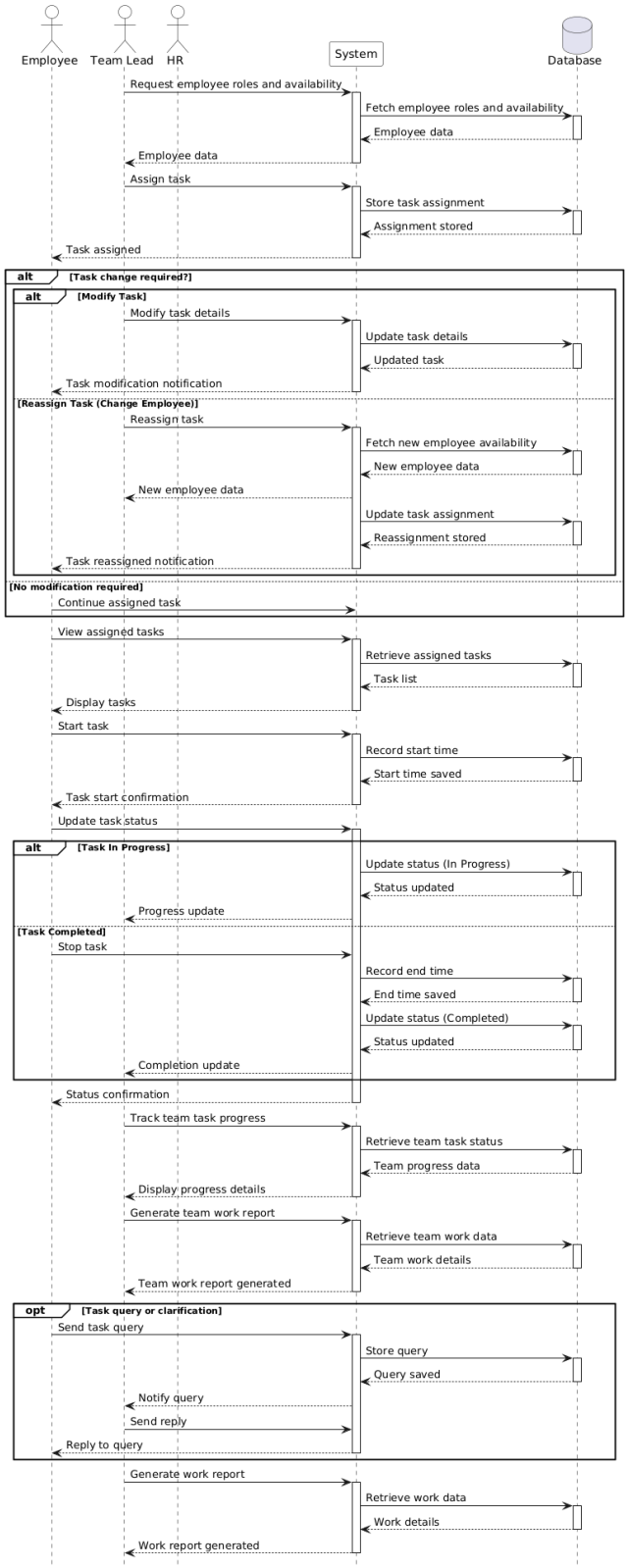


Manage Policy:

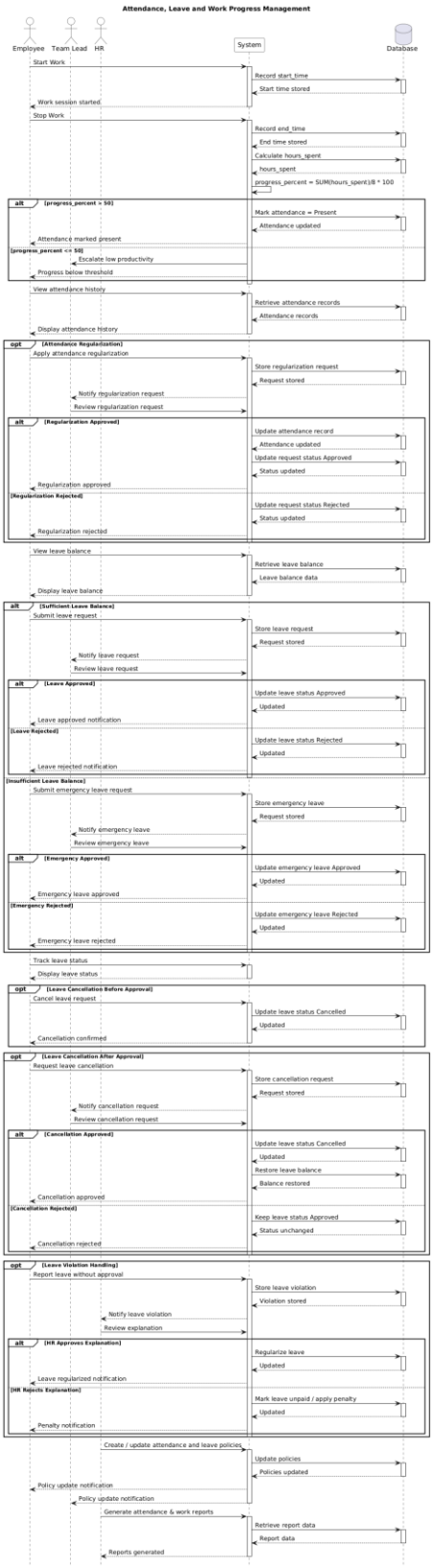


6.3 Sequence Diagrams

Sequence_work:



Sequence Attendance and Leave:



7 Appendices

Hardware Requirements

- Processor: Dual-core processor or higher
- RAM: 4 GB minimum
- Storage: 10 GB free disk space
- Display: Standard monitor
- Network: Internet connectivity
- Input Devices: Keyboard and Mouse

Database Requirements

The database stores information related to employees, attendance records, leave details, work allocation, and task progress.

Key Data Entities

- Employee
- Attendance
- Attendance Regularization
- Leave
- Leave Balance
- Task
- Work Allocation
- Team Lead
- HR

Data Relationships

- An employee can have multiple attendance records.
- An employee can submit multiple leave requests.
- Each leave request is linked to one employee and one approver (Team Lead or HR).
- A team lead can allocate multiple tasks to employees.
- Each task is assigned to one employee.
- Attendance regularization requests are linked to attendance records.

8 Future Scope

- Add features to better understand attendance and work patterns.
- Make the system available on mobile devices and cloud platforms.
- Improve reports to help team leads and HR make better decisions.

9 Conclusion

This system streamlines employee attendance tracking, leave management, and work assignment processes, thereby improving efficiency and reducing manual errors. By providing role-based access for employees, team leads, and HR, the platform ensures accurate record keeping, better coordination, and smooth workforce management operations.